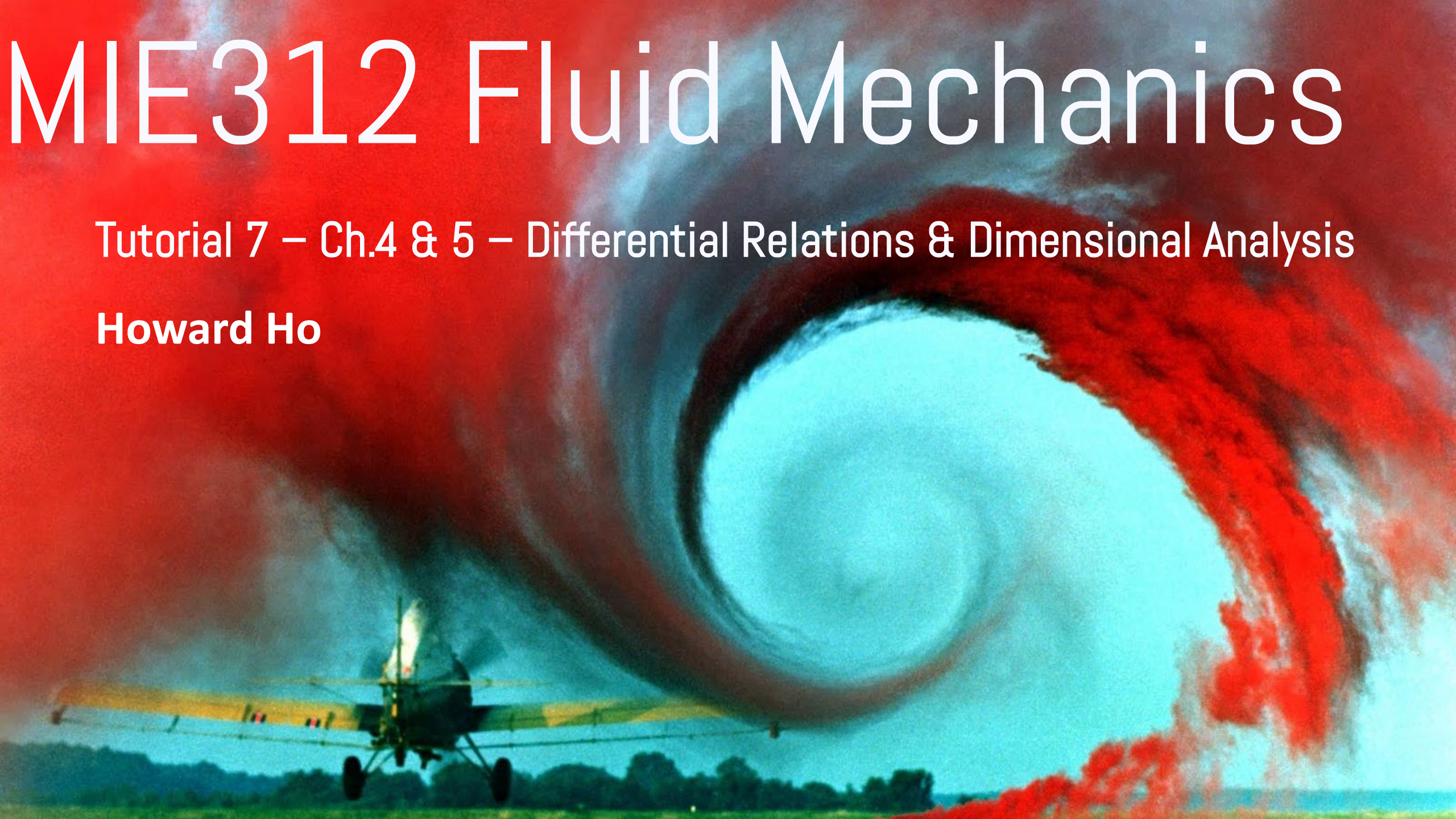
MIE312 Fluid Mechanics

Tutorial 7 – Ch.4 & 5 – Differential Relations & Dimensional Analysis

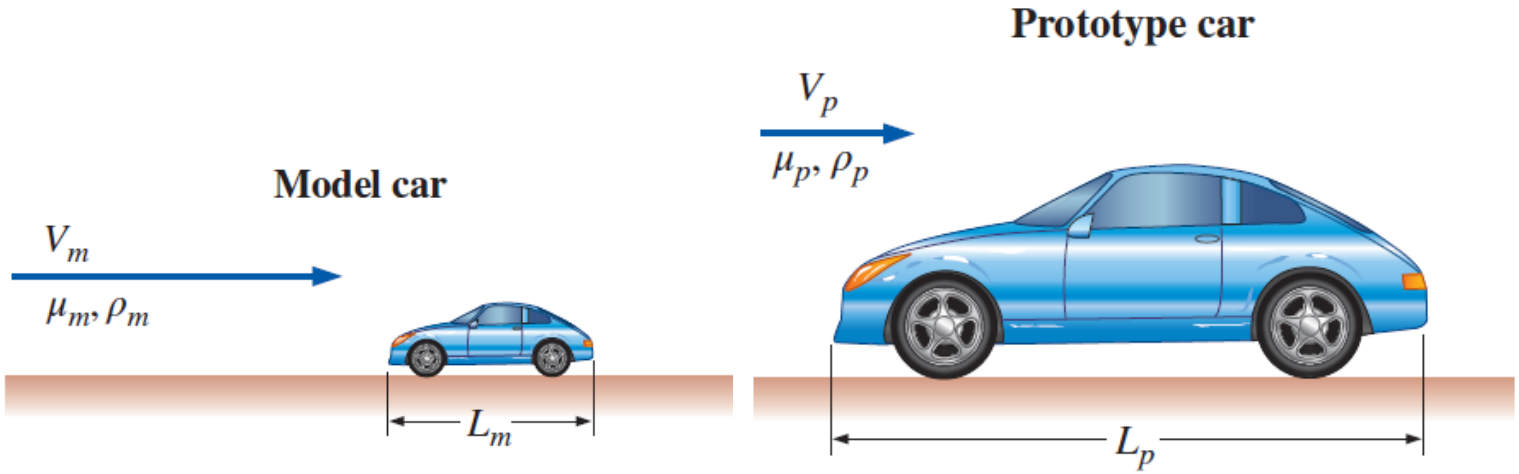
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Learning Objective

- Applying differential equations on a fluids problem
- Apply the Buckingham Pi Theorem to reduce complex problems into dimensionless groups
- Use dimensional analysis to scale and model fluid systems*

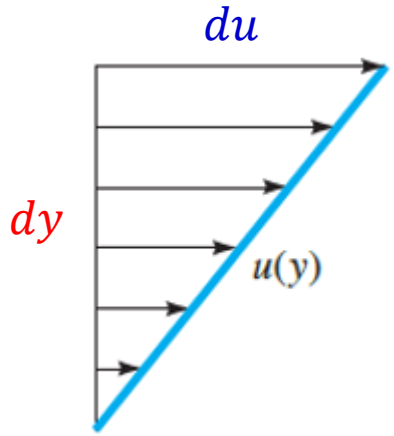


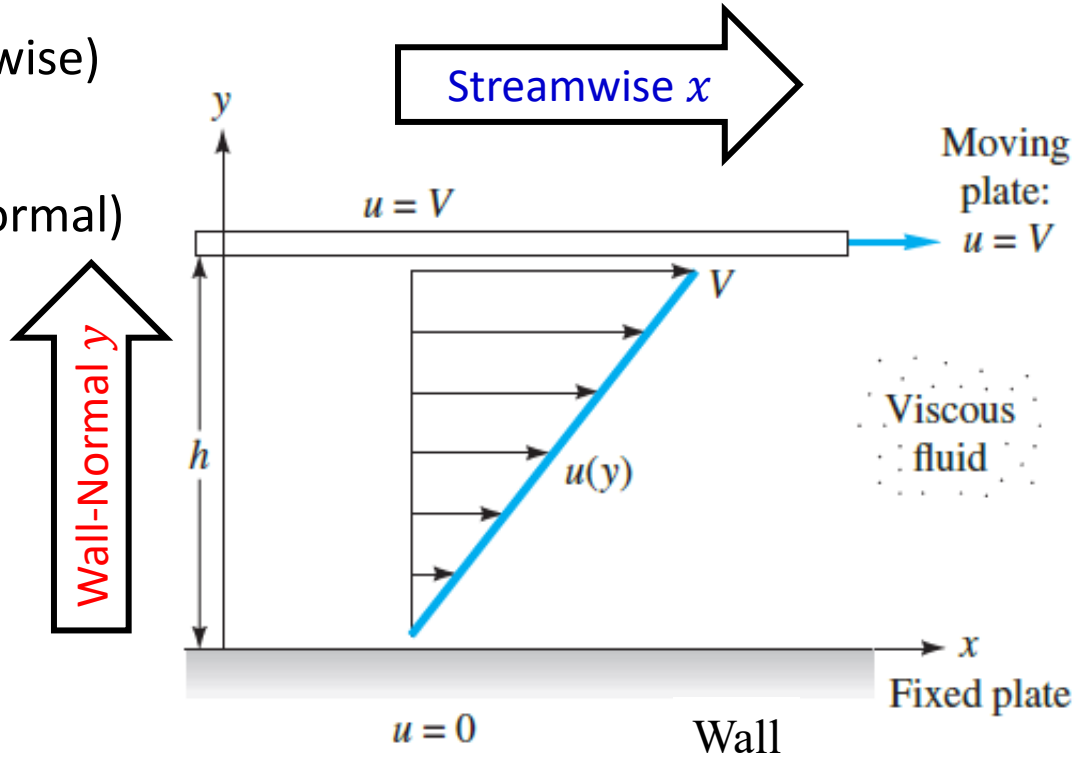


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Differential Equations

- The main focus is on velocity gradient,
  - $\frac{du}{dy}$: Streamwise velocity u changes in the y-direction (wall-normal)
  - $\frac{dv}{dx}$: Wall-normal velocity v changes in the x-direction (streamwise)
  - $\frac{du}{dx}$: Streamwise velocity u changes in the x-direction (streamwise)
  - $\frac{dv}{dy}$: Wall-normal velocity v changes in the y-direction (wall-normal)





13. DIFFERENTIAL FLUID MOTION

2-D field: $\mathbf{V} = u(x, y)\mathbf{i} + v(x, y)\mathbf{j}$

Normal strain: $\dot{\epsilon}_{xx} = \frac{\partial u}{\partial x}$; $\dot{\epsilon}_{yy} = \frac{\partial v}{\partial y}$

Angular deformation: $\dot{\gamma}_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$

Vorticity: $\boldsymbol{\zeta} = \nabla \times \mathbf{V}$; $\zeta_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$

Element rotation: $\boldsymbol{\Omega} = \frac{1}{2}\boldsymbol{\zeta}$

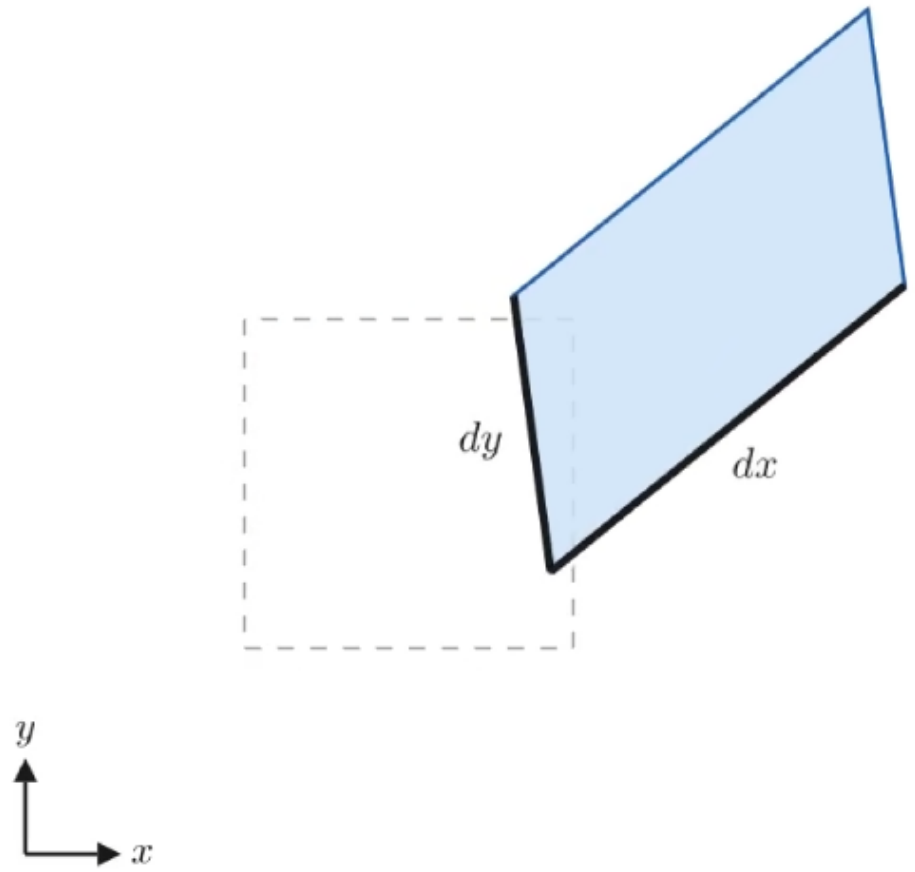
2-D incompressible: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

No slip at a wall: $\mathbf{V}_{fluid} = \mathbf{V}_{wall}$; use the stated profile form and wall conditions to determine constants

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Fluid Motions

Depending on the flow, sometimes all four can happen at the same time



translation

+ rotation

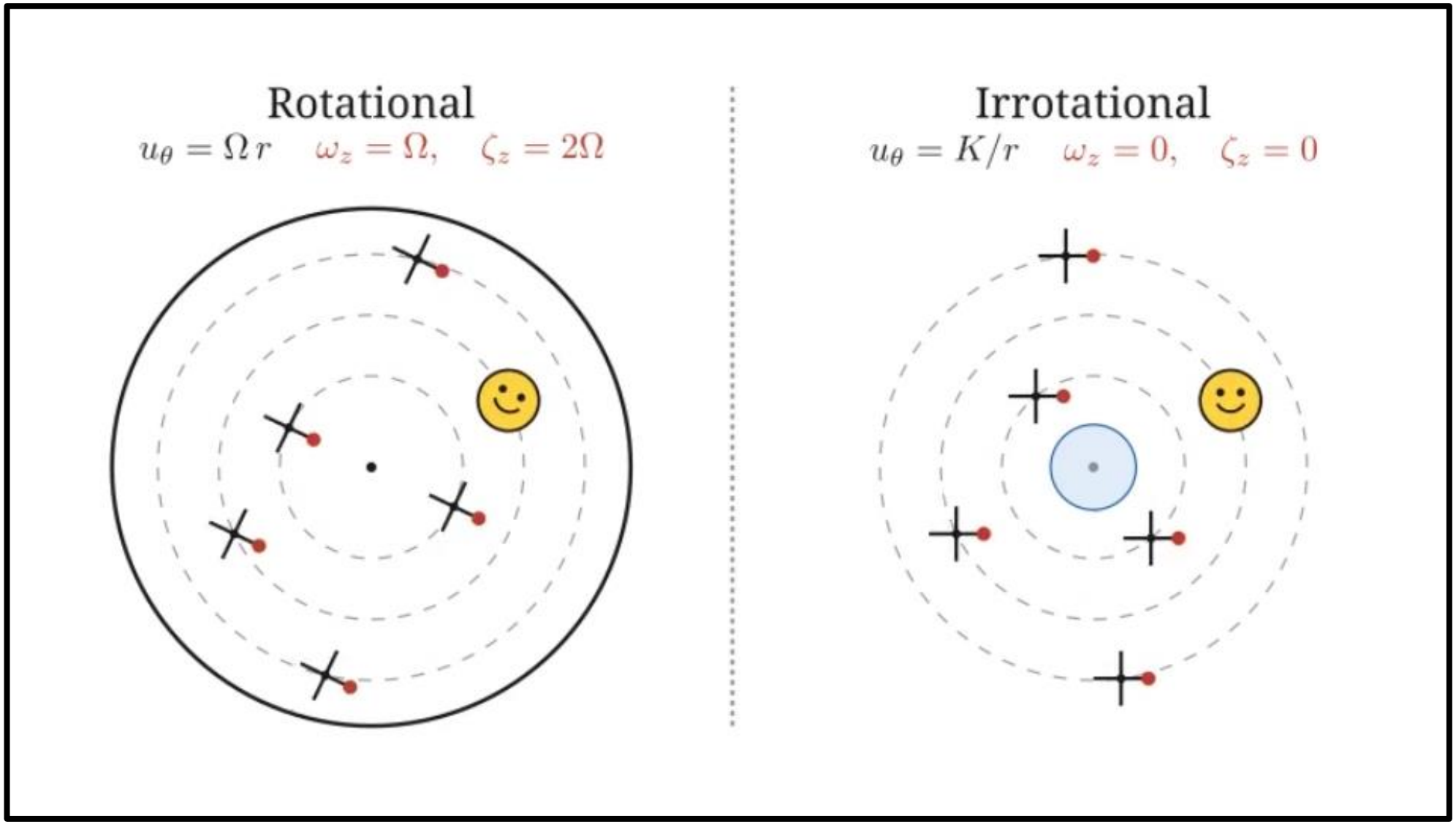
+ linear strain

+ shear strain

Link to video: **[here]**

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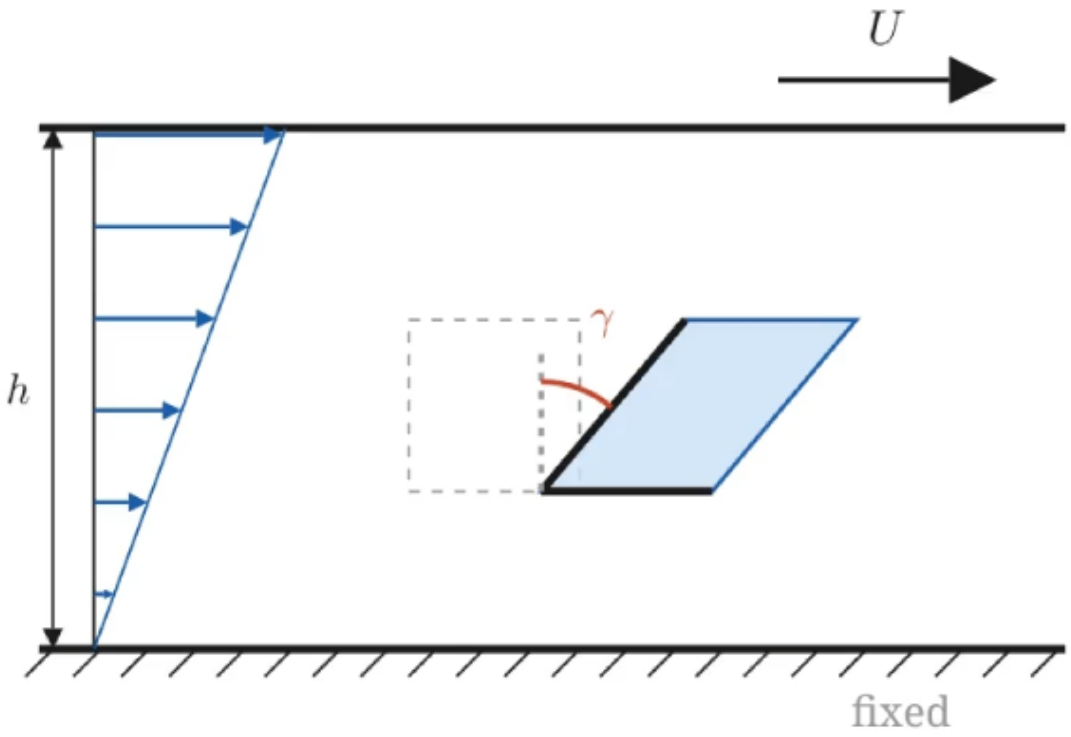
Rotation (Vorticity)



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Shear

Straight streamlines, yet rotational (unlike the curved free vortex)

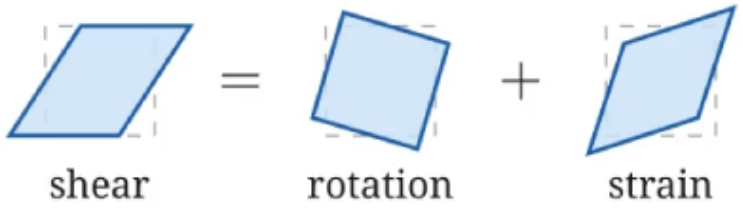


$$u = U\frac{y}{h}, \qquad v = 0$$

$$\zeta_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -\frac{U}{h}$$

$$\omega_z = -\frac{U}{2h} \quad \text{(clockwise)}$$

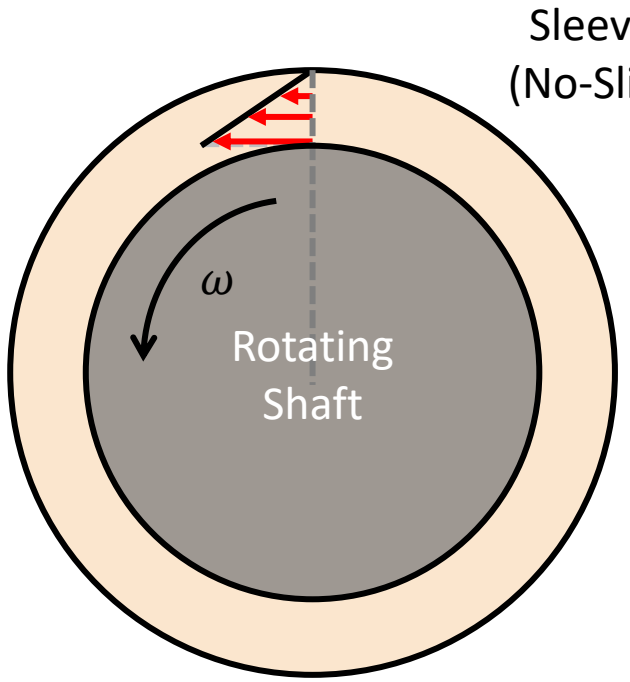
$$\varepsilon_{xy} = \frac{1}{2}\frac{\partial u}{\partial y} = \frac{U}{2h}$$



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Problem 1

Figure on the right shows the cross section of a hydrostatic sleeve bearing on a lathe. Oil resides between the fixed sleeve and the rotating shaft. The velocity profile of the fluid is linear and can be treated as straight channel flow, derive an expression for: (a) Angular deformation $\dot{\gamma}_{xy}$; (b) Vorticity ξ_z



13. DIFFERENTIAL FLUID MOTION

2-D field: $\mathbf{V} = u(x,y)\mathbf{i} + v(x,y)\mathbf{j}$

Normal strain: $\dot{\epsilon}_{xx} = \frac{\partial u}{\partial x}$; $\dot{\epsilon}_{yy} = \frac{\partial v}{\partial y}$

Angular deformation: $\dot{\gamma}_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$

Vorticity: $\boldsymbol{\zeta} = \nabla \times \mathbf{V}$; $\zeta_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$

Element rotation: $\boldsymbol{\Omega} = \frac{1}{2}\boldsymbol{\zeta}$

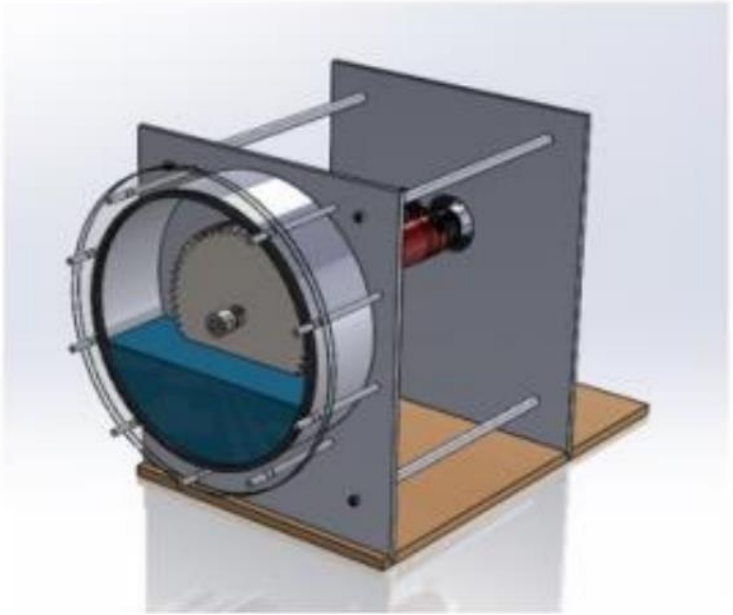
2-D incompressible: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

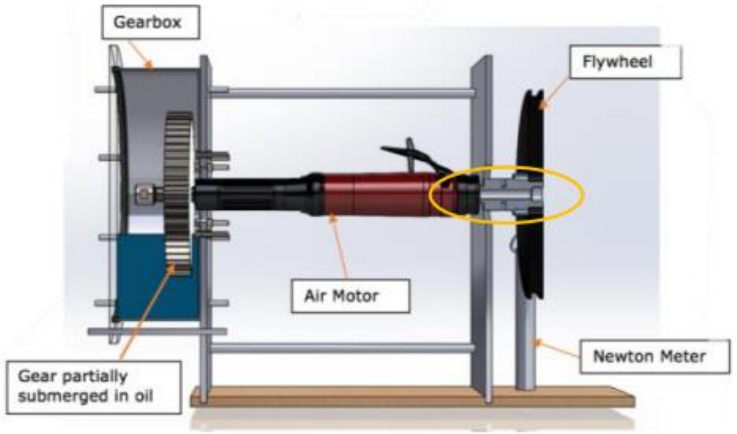
No slip at a wall: $\mathbf{V}_{fluid} = \mathbf{V}_{wall}$; use the stated profile form and wall conditions to determine constants

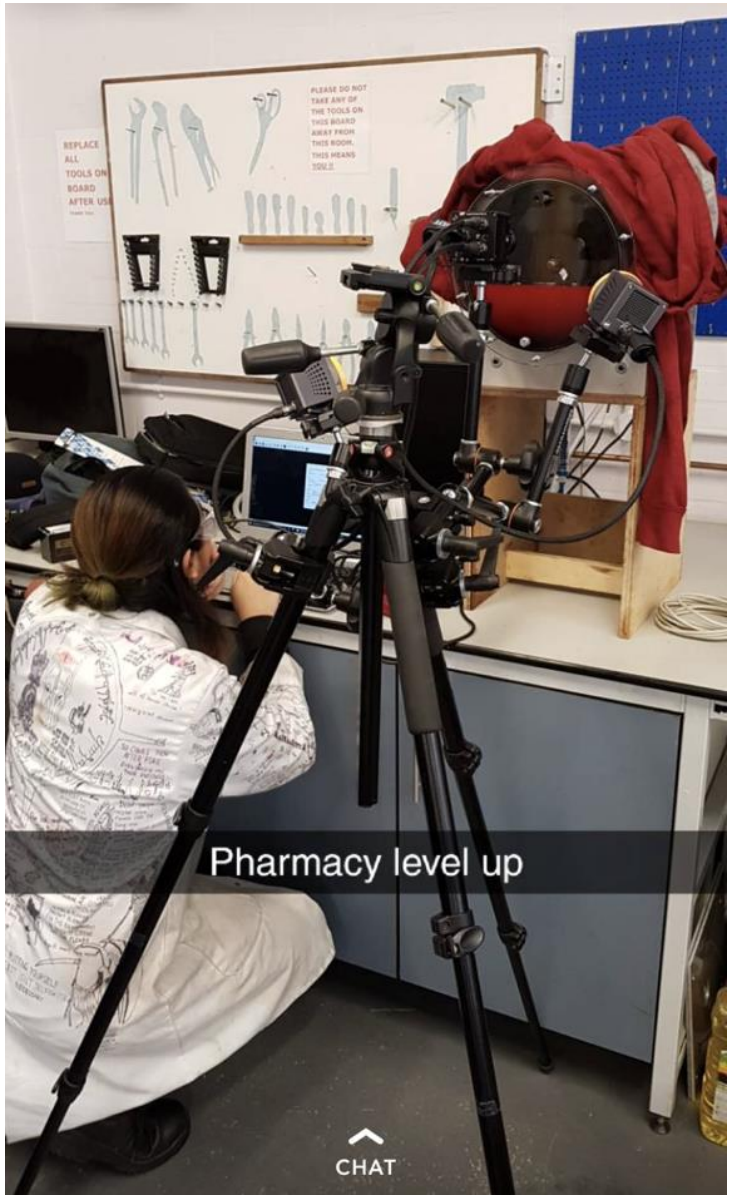
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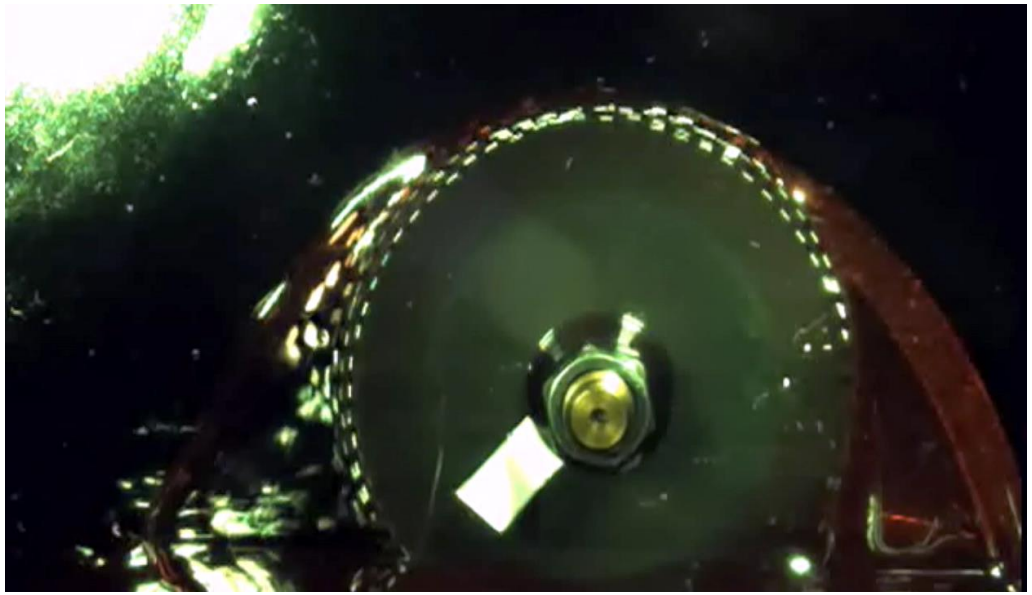
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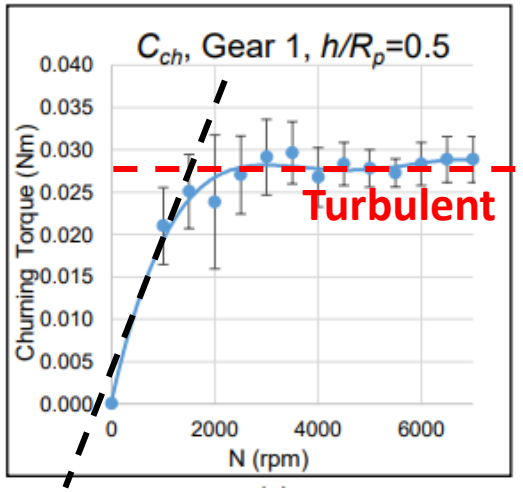
My Capstone Project

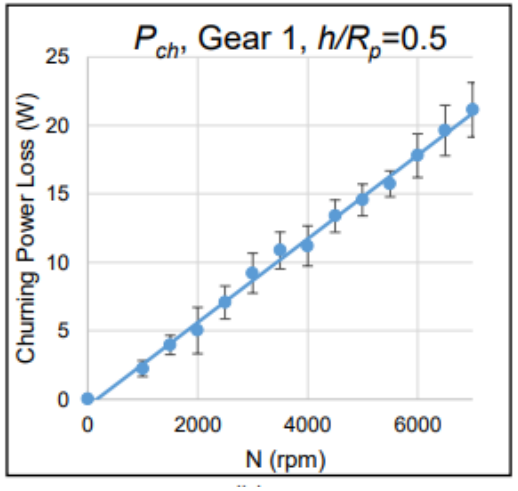












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Appendix D – Prove for Equation (2.17)

A table of Dimensions of standard thermofluid quantities was used for completing this task.

$$C_{ch} = f(m, D_p, b, \mu, \rho, h, V_o, g, \Omega)$$

$C_{ch} = ML^2T^{-2}$
$m = L$
$D_p = L$
$b = L$
$\mu = ML^{-1}T^{-1}$
$\rho = ML^{-3}$
$h = L$
$V_o = L^3$
$g = LT^{-2}$
$\Omega = T^{-1}$

10 variables, 3 dimensions

m=7 (Number of steps)

Repeating variables chosen $= \Omega, \rho, D_p$

$$[\Pi_1] = [M^0L^0T^0]$$
$$= [C_{ch}][\Omega]^a[\rho]^b[D_p]^c$$
$$= (ML^2T^{-2})(T^{-1})^a(ML^{-3})^b(L)^c$$

$M: 0 = 1 + b \rightarrow b = -1$
$L: 0 = 2 - 3b + c \rightarrow c = 3b - 2 = -5$
$T: 0 = -2 - a \rightarrow a = -2$

$$\therefore \Pi_1 = \frac{C_{ch}}{\Omega^2\rho^1 D_p^5} \Big\} \; C_m(\textit{dimensionlesss torque})$$

Π2=M0L0T0
=mΩdpeD*pf*
=LT−1*d*ML−3e*L*f

$M: 0 = e \rightarrow e = 0$
$L: 0 = 1 - 3e + f \rightarrow f = -1$
$T: 0 = -d \rightarrow d = 0$

$$\therefore \Pi_2 = \frac{m}{D_p}$$

$$[\Pi_3] = [M^0L^0T^0]$$
$$= [b][\Omega]^g[\rho]^h[D_p]^i$$
$$= (L)(T^{-1})^g(ML^{-3})^h(L)^i$$

$M: 0 = h \rightarrow h = 0$
$L: 0 = 1 - 3h + i \rightarrow i = -1$
$T: 0 = -g \rightarrow g = 0$

$$\therefore \Pi_3 = \frac{b}{D_p}$$

$$[\Pi_4] = [M^0L^0T^0]$$
$$= [\mu][\Omega]^a[\rho]^b[D_p]^c$$
$$= (ML^{-1}T^{-1})(T^{-1})^a(ML^{-3})^b(L)^c$$

$M: 0 = 1 + b \rightarrow b = -1$
$L: 0 = -1 - 3b + c \rightarrow c = 3b + 1 = -2$
$T: 0 = -1 - a \rightarrow a = -1$

$$\therefore \Pi_4 = \frac{\mu}{\Omega\rho D_p^2} = \frac{1}{Re} = Re$$

$$\textit{Where } \Omega = \frac{U}{r}$$

$$[\Pi_5] = [M^0L^0T^0]$$
$$= [V_o][\Omega]^d[\rho]^e[D_p]^f$$
$$= (L^3)(T^{-1})^d(ML^{-3})^e(L)^f$$

$M: 0 = e \rightarrow e = 0$
$L: 0 = 3 - 3e + f \rightarrow f = -3$
$T: 0 = -d \rightarrow d = 0$

$$\therefore \Pi_5 = \frac{V_o}{D_p^3}$$

$$[\Pi_6] = [M^0L^0T^0]$$
$$= [g][\Omega]^g[\rho]^h[D_p]^i$$
$$= (LT^{-2})(T^{-1})^g(ML^{-3})^h(L)^i$$

$M: 0 = h \rightarrow h = 0$
$L: 0 = 1 - 3h + i \rightarrow i = -1$
$T: 0 = -2 - g \rightarrow g = -2$

$$\therefore \Pi_6 = \frac{g}{\Omega^2 D_p} = \frac{g}{\left(\frac{U}{D_p}\right)^2 D_p} = \frac{1}{Fr} = Fr$$

$$[\Pi_7] = [M^0L^0T^0]$$
$$= [h][\Omega]^j[\rho]^k[D_p]^l$$
$$= (L)(T^{-1})^j(ML^{-3})^k(D_p)^l$$

$M: 0 = k \rightarrow k = 0$
$L: 0 = 1 - 3k + l \rightarrow l = -1$
$T: 0 = -j \rightarrow j = 0$

$$\therefore \Pi_7 = \frac{h}{Dp}$$

$$\therefore \frac{C_{ch}}{\Omega^2\rho D_p^5} = f\left(\frac{m}{D_p}, \frac{b}{D_p}, Re, \frac{V_o}{D_p^3}, Fr, \frac{h}{D_p}\right)$$

And finally,

$$C_m = \varphi_1\left(\frac{m}{D_p}\right)^{\varphi_2}\left(\frac{b}{D_p}\right)^{\varphi_3}\left(\frac{h}{D_p}\right)^{\varphi_4}\left(\frac{V_o}{D_p^3}\right)^{\varphi_5} Re^{\varphi_6} Fr^{\varphi_7}$$

Changenet et al[3] suggested the C_m can be expressed as:

if $\gamma < 750$ m/s^2 and $Re_\cdot < 4000$

$$C_m = 1.366\left(\frac{h}{D_p}\right)^{0.45}\left(\frac{V_0}{D_p^3}\right)^{0.1} Fr^{-0.6} Re^{-0.21}\left(\frac{b}{R_p}\right)^{0.21} \quad (1)$$

if $\gamma < 750$ m/s^2 and $Re_\cdot > 4000$

$$C_m = 0.239\left(\frac{h}{D_p}\right)^{0.45}\left(\frac{V_0}{D_p^3}\right)^{0.1} Fr^{-0.6}\left(\frac{b}{R_p}\right)^{0.21} \quad (2)$$

You can read the presentation I gave to the industry sponsor here: Slides

For those who wanted some practice/ challenge, this is from my capstone project, slight rendering issue with Π_2 but rest are fine. **[This is very over-kill, you won't run into something as lengthy as this in MIE 312.]**

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Fundamental Units

- There are 7 fundamental dimensions but we only use 3 within the scope of this course (thankfully)
  - Length (L)
  - Mass (M)
  - Time (T)
- Sometimes Mass (M) is replaced with Force (F)
  - Length (L)
  - Force (F)
  - Time (T)
- Note Force has a dimension of $[\frac{ML}{T^2}]$

Quantity	Symbol	Dimensions	
		$MLT\Theta$	$FLT\Theta$
Length	L	L	L
Area	A	L^2	L^2
Volume	$\mathcal{V}$	L^3	L^3
Velocity	V	LT^{-1}	LT^{-1}
Acceleration	dV/dt	LT^{-2}	LT^{-2}
Speed of sound	a	LT^{-1}	LT^{-1}
Volume flow	Q	L^3T^{-1}	L^3T^{-1}
Mass flow	$\dot{m}$	MT^{-1}	FTL^{-1}
Pressure, stress	p, σ, τ	$ML^{-1}T^{-2}$	FL^{-2}
Strain rate	$\dot{\varepsilon}$	T^{-1}	T^{-1}
Angle	θ	None	None
Angular velocity	ω, Ω	T^{-1}	T^{-1}
Viscosity	μ	$ML^{-1}T^{-1}$	FTL^{-2}
Kinematic viscosity	ν	L^2T^{-1}	L^2T^{-1}
Surface tension	Υ	MT^{-2}	FL^{-1}
Force	F	MLT^{-2}	F
Moment, torque	M	ML^2T^{-2}	FL
Power	P	ML^2T^{-3}	FLT^{-1}
Work, energy	W, E	ML^2T^{-2}	FL
Density	ρ	ML^{-3}	FT^2L^{-4}

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Steps To Find Πs (Long version)*

Step 1 Write the function and count variables:

$$F = f(L, U, \rho, \mu) \quad \text{there are five variables } (n = 5)$$

Step 2 List dimensions of each variable. From Table 5.1

F	L	U	ρ	μ
$\{MLT^{-2}\}$	$\{L\}$	$\{LT^{-1}\}$	$\{ML^{-3}\}$	$\{ML^{-1}T^{-1}\}$

Step 3 Find j. No variable contains the dimension Θ, and so j is less than or equal to 3 (MLT). We inspect the list and see that L, U, and ρ cannot form a pi group because only ρ contains mass and only U contains time. Therefore j does equal 3, and $n - j = 5 - 3 = 2 = k$. The pi theorem guarantees for this problem that there will be exactly two independent dimensionless groups.

Step 4 Select repeating j variables. The group L, U, ρ we found in step 3 will do fine.

Step 5 Combine L, U, ρ with one additional variable, in sequence, to find the two pi products.

First add force to find Π_1. You may select *any* exponent on this additional term as you please, to place it in the numerator or denominator to any power. Since F is the output, or dependent, variable, we select it to appear to the first power in the numerator:

$$\Pi_1 = L^a U^b \rho^c F = (L)^a (LT^{-1})^b (ML^{-3})^c (MLT^{-2}) = M^0 L^0 T^0$$

Equate exponents:

Length: $a + b - 3c + 1 = 0$

Mass: $c + 1 = 0$

Time: $-b \quad -2 = 0$

We can solve explicitly for

$$a = -2 \qquad b = -2 \qquad c = -1$$

Therefore $$\Pi_1 = L^{-2} U^{-2} \rho^{-1} F = \frac{F}{\rho U^2 L^2} = C_F \qquad \textit{Ans.}$$

Step 5 (Cont.)

Finally, add viscosity to L, U, and ρ to find Π_2. Select any power you like for viscosity. By hindsight and custom, we select the power -1 to place it in the denominator:

$$\Pi_2 = L^a U^b \rho^c \mu^{-1} = L^a (LT^{-1})^b (ML^{-3})^c (ML^{-1}T^{-1})^{-1} = M^0 L^0 T^0$$

Equate exponents:

Length: $a + b - 3c + 1 = 0$

Mass: $c - 1 = 0$

Time: $-b \quad + 1 = 0$

from which we find

$$a = b = c = 1$$

Therefore $$\Pi_2 = L^1 U^1 \rho^1 \mu^{-1} = \frac{\rho UL}{\mu} = \text{Re} \qquad \textit{Ans.}$$

Step 6 We know we are finished; this is the second and last pi group. The theorem guarantees that the functional relationship must be of the equivalent form

$$\frac{F}{\rho U^2 L^2} = g\left(\frac{\rho UL}{\mu}\right) \qquad \textit{Ans.}$$

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Short Version

A wise choice for repeating parameters for most fluid flow problems are a **length**, a **velocity** and a **mass** or **density**.

6-Steps:

1) n = number of variables

2) List dimensions for each n

3) Find $k = n - j$ (j = number of dimensions, usually $j = 3$)

4) Choose repeating j (normally provided in exams)

5) Construct Πs

6) Write combined equation and check dimensions

1) no. of n
2) List dimensions
3) $k = n - j$
4) Choose j
5) Find Πs
6) Combine

16. NOMENCLATURE & DIMS

m	Mass	kg	M
A, D	Area, Dia	m^2, m	L^2, L
I	Second Moment of Area	m^4	L^4
v, a	Velo, Accel	$m/s, m/s^2$	LT^{-1}, LT^{-2}
F, T	Force, Torque	N, Nm	$ML/T^2, ML^2/T^2$
$P, \dot{W}$	Pressure, Power	Pa, W	$M/LT^2, ML^2/T^3$
γ, ρ	Sp Weight, Dens	$N/m^3, kg/m^3$	$M/L^2T^2, M/L^3$
μ, ν	Viscosity	$Pa \cdot s, m^2/s$	$M/LT, L^2/T$
τ	Shear Stress	Pa	$ML^{-1}T^{-2}$
Υ	Surf. Tension	N/m	MT^{-2}

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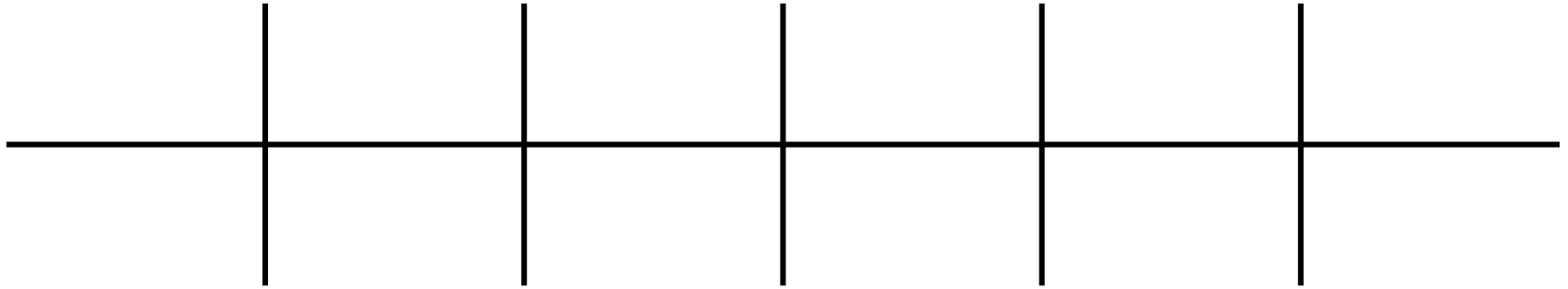
Question 2

A fixed cylinder of diameter D and length L, immersed in a stream flowing normal to its axis at velocity U, will experience zero average lift. However, if the cylinder is rotating at angular velocity Ω, a lift force F will arise. The fluid density ρ is important, but viscosity is secondary and can be neglected. Formulate this lift behavior as a dimensionless function. Hint: Use D, U, ρ as repeating variables.

16. NOMENCLATURE & DIMS

m	Mass	kg	M
A, D	Area, Dia	m^2, m	L^2, L
I	Second Moment of Area	m^4	L^4
V, a	Velo, Accel	$m/s, m/s^2$	LT^{-1}, LT^{-2}
F, T	Force, Torque	N, Nm	MLT^{-2}, ML^2T^{-2}
$P, \dot{W}$	Pressure, Power	Pa, W	$ML^{-1}T^{-2}, ML^2T^{-3}$
γ, ρ	Sp Weight, Dens	$N/m^3, kg/m^3$	$ML^{-2}T^{-2}, ML^{-3}$
μ, ν	Viscosity	$Pa \cdot s, m^2/s$	$ML^{-1}T^{-1}, L^2T^{-1}$
τ	Shear Stress	Pa	$ML^{-1}T^{-2}$
Υ	Surf. Tension	N/m	MT^{-2}

1) no. of n
2) List dimensions
3) $k = n - j$
4) Choose j
5) Find Πs
6) Combine



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16. NOMENCLATURE & DIMS

m	Mass	kg	M
A, D	Area, Dia	m^2, m	L^2, L
I	Second Moment of Area	m^4	L^4
V, a	Velo, Accel	$m/s, m/s^2$	LT^{-1}, LT^{-2}
F, T	Force, Torque	N, Nm	MLT^{-2}, ML^2T^{-2}
$P, \dot{W}$	Pressure, Power	Pa, W	$ML^{-1}T^{-2}, ML^2T^{-3}$
γ, ρ	Sp Weight, Dens	$N/m^3, kg/m^3$	$ML^{-2}T^{-2}, ML^{-3}$
μ, ν	Viscosity	$Pa \cdot s, m^2/s$	$ML^{-1}T^{-1}, L^2T^{-1}$
τ	Shear Stress	Pa	$ML^{-1}T^{-2}$
Υ	Surf. Tension	N/m	MT^{-2}

1) no. of n
2) List dimensions
3) $k = n - j$
4) Choose j
5) Find Πs
6) Combine

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16. NOMENCLATURE & DIMS

m	Mass	kg	M
A, D	Area, Dia	m^2, m	L^2, L
I	Second Moment of Area	m^4	L^4
V, a	Velo, Accel	$m/s, m/s^2$	LT^{-1}, LT^{-2}
F, T	Force, Torque	N, Nm	MLT^{-2}, ML^2T^{-2}
$P, \dot{W}$	Pressure, Power	Pa, W	$ML^{-1}T^{-2}, ML^2T^{-3}$
γ, ρ	Sp Weight, Dens	$N/m^3, kg/m^3$	$ML^{-2}T^{-2}, ML^{-3}$
μ, ν	Viscosity	$Pa \cdot s, m^2/s$	$ML^{-1}T^{-1}, L^2T^{-1}$
τ	Shear Stress	Pa	$ML^{-1}T^{-2}$
Υ	Surf. Tension	N/m	MT^{-2}

1) no. of n
2) List dimensions
3) $k = n - j$
4) Choose j
5) Find Πs
6) Combine

12. DIMENSIONAL ANALYSIS

Pi Theorem: number of Π groups $= n - r$, where n is the number of variables and r is the dimensional-matrix rank. Repeating variables must be dimensionally independent.

$Re = \frac{\rho V L}{\mu}$; $Fr = \frac{V}{\sqrt{gL}}$; $Ma = \frac{V}{a}$; $We = \frac{\rho V^2 L}{\Upsilon}$

$Eu = \frac{\Delta P}{\rho V^2}$; $St = \frac{\omega L}{V}$; $Ca = \frac{\mu V}{\Upsilon}$

$\Pi_\tau = \frac{\tau_w}{\rho V^2}$; $f_D = 8\Pi_\tau$; laminar $\Pi_\tau = 8/Re$;

Blasius $\Pi_\tau \approx 0.0396 Re^{-1/4}$; $\sigma_{cav} = \frac{P - P_v}{\frac{1}{2}\rho V^2}$; $C_p = \frac{P - P_\infty}{\frac{1}{2}\rho V_\infty^2}$

Similarity: $\Pi_{model} = \Pi_{prototype}$

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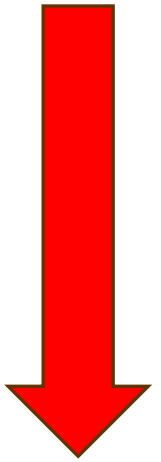
Similarity (Scaling)*

Three necessary conditions for complete similarity between a model and a prototype.

(1) Geometric similarity—the model must be the same shape as the prototype but may be scaled by some constant scale factor.

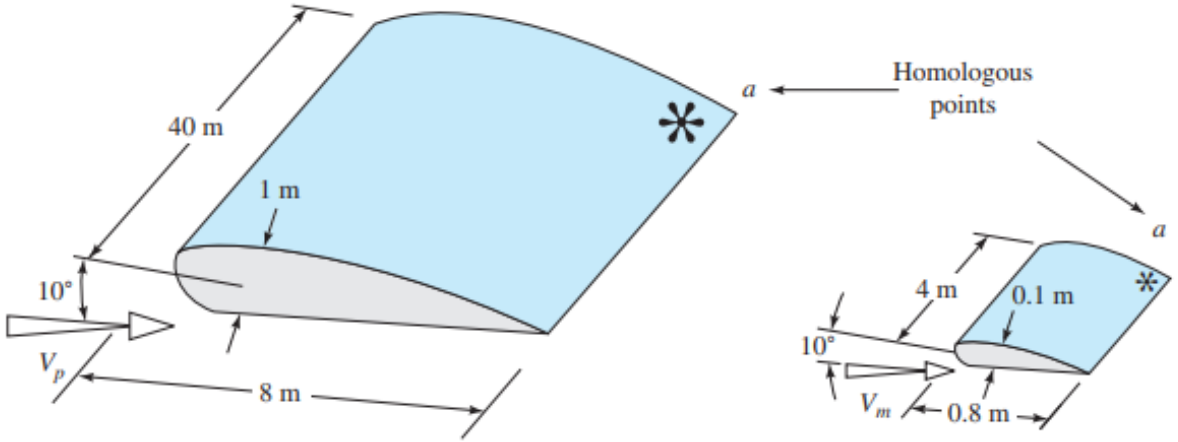
(2) Kinematic similarity—the velocity at any point in the model flow must be proportional (by a constant scale factor) to the velocity at the corresponding point in the prototype flow.

(3) Dynamic similarity—When all *forces* in the model flow scale by a constant factor to corresponding forces in the prototype flow (*force-scale* equivalence).



Increasing Difficulties

1. The model nose radius must be one-tenth as large.
2. The model surface roughness must be one-tenth as large.



True Complete Similarity only occurs when all three conditions above are met (and it's not an easy process once you consider dynamic similarity)

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Similarity (Scaling)*

(2) Kinematic similarity—the velocity at any point in the model flow must be proportional (by a constant scale factor) to the velocity at the corresponding point in the prototype flow. Sometimes the test are done in different fluids for convenience, and we want to match dimensionless parameters between the model and the prototype:

$$\Pi_{model} = \Pi_{prototype}$$

For example (but not limited to):

$$Re_{prototype} = Re_{model}$$

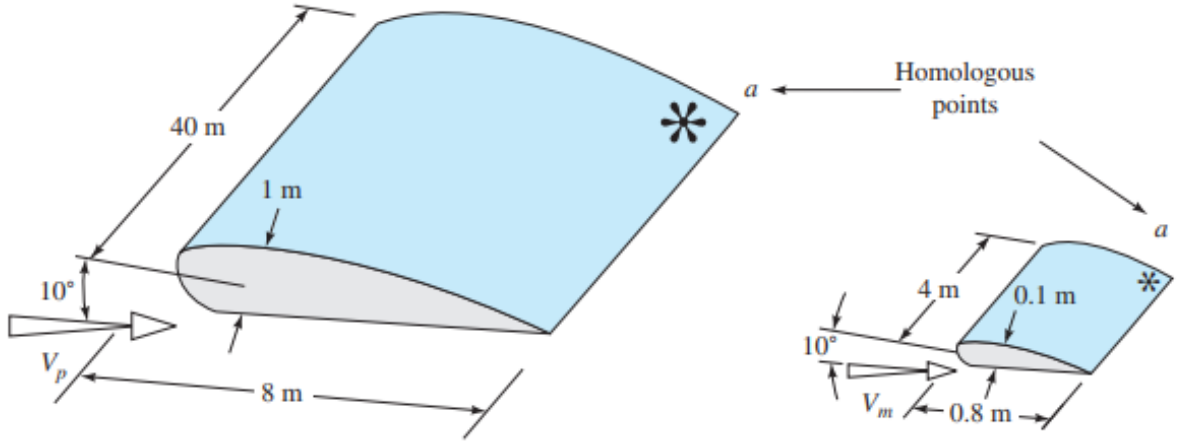
$$\frac{\rho_p V_p L_p}{\mu_p} = \frac{\rho_m V_m L_m}{\mu_m}$$

$$Ma_{prototype} = Ma_{model}$$

$$\frac{V_p}{a_p} = \frac{V_m}{a_m}$$



Reynolds number is one of the most important dimensionless number for characterising flow



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Question 3*

P5.81 An airplane, of overall length 17 m, is designed to fly at 680 m/s at 8000-m standard altitude. A one-thirtieth-scale model is to be tested in a pressurized helium wind tunnel at 20°C. What is the appropriate tunnel pressure in atm? Even at this (high) pressure, exact dynamic similarity is not achieved. Why?



air:
$\rho = 0.525\ kg/m^3$
$\mu = 1.53e^{-5}\ kg/ms$
$a = 308\ m/s$
$R = 2077\ J/kg°K$

helium:
$\mu = 1.97e^{-5}\ kg/ms$
$a = 1005\ m/s$

Kinematic Similarity:

$Ma_{prototype} = Ma_{model}$

$\frac{V_p}{a_p} = \frac{V_m}{a_m}$

- Reynolds number (Re)
- Mach number (Ma)

$Ma_p = \frac{680}{308} = 2.21$ (prototype)

we want matching Ma.

$Ma_m = \frac{V_m}{1005} = 2.21$ (model)

$\therefore V_m = 2219\ m/s$

Repeat for Re.

$Re_p = \frac{\rho_p V_p L_p}{\mu_p} = \frac{0.525(680)(17)}{1.53e^{-5}} = 3.96e^8$

The model is 1/30 scale

$Re_m = 3.96e^8 = \frac{\rho_{He} V_m L_m}{\mu_{He}} = \frac{\rho_{He}(2219)(17/30)}{1.97e^{-5}}$

$\therefore \rho_{He} = 6.21\ kg/m^3$

Ideal Gas Law: $P = \rho R T$

$P = 6.21(2077)(20)$

$P = 3.78\ MPa$

$Re_{prototype} = Re_{model}$

$\frac{\rho_p V_p L_p}{\mu_p} = \frac{\rho_m V_m L_m}{\mu_m}$

$Ma_{prototype} = Ma_{model}$

$\frac{V_p}{a_p} = \frac{V_m}{a_m}$

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Question 3

P5.81 An airplane, of overall length 17 m, is designed to fly at 680 m/s at 8000-m standard altitude. A one-thirtieth-scale model is to be tested in a pressurized helium wind tunnel at 20°C. What is the appropriate tunnel pressure in atm? Even at this (high) pressure, exact dynamic similarity is not achieved. Why?

The last part of the question is somewhat beyond the scope of this course since it involves thermoldynamics.

Solution: For air at 8000-m standard altitude (Table A-6), take $\rho = 0.525$ kg/m^3, $\mu = 1.53E{-}5$ kg/m·s, and sound speed $a = 308$ m/s. For helium at 20°C (Table A-4), take gas constant R = 2077 J/(kg·°K), $\mu = 1.97E{-}5$ kg/m·s, and $a = 1005$ m/s. For similarity at this supersonic speed, we must match both the Mach and Reynolds numbers.

$$\mathrm{Ma_p} = \frac{680}{308} = 2.21 = \mathrm{Ma_m} = \frac{V_m}{1005}, \quad \text{solve for } V_{model} \approx 2219 \frac{m}{s}$$

$$\mathrm{Re_p} = \frac{\rho V L}{\mu}\Big|_p = \frac{0.525(680)(17)}{1.53E{-}5} = 3.96E8 = \mathrm{Re_m} = \frac{\rho_{He}(2219)(17/30)}{1.97E{-}5}$$

$$\text{Solve for } \rho_{He} \approx 6.21 \text{ kg/m}^3 = \frac{p}{RT} = \frac{p_{He}}{(2077)(293)},$$

$$\text{or } \mathbf{p_{He} \approx 3.78 \text{ MPa} = 37.3 \text{ atm}} \quad \textit{Ans.}$$

Even with Ma and Re matched, true dynamic similarity is not achieved, because the specific heat ratio of helium, $k \approx 1.66$, is not equal to $k_{air} \approx 1.40$.

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Practice Problems

P5.14 Flow in a pipe is often measured with an orifice plate, as in Fig. P5.14. The volume flow Q is a function of the pressure drop Δp across the plate, the fluid density ρ, the pipe diameter D, and the orifice diameter d. Rewrite this functional relationship in dimensionless form.

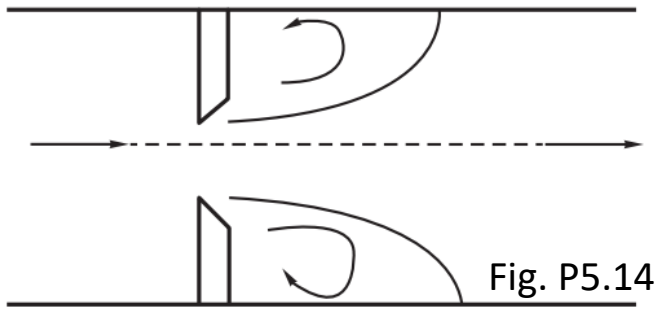


Fig. P5.14

P5.5 An automobile has a characteristic length and area of 8 ft and 60 ft^2, respectively. When tested in sea-level standard air, it has the following measured drag force versus speed:

V, mi/h	20	40	60
Drag, lbf	31	115	249

The same car travels in Colorado at 65 mi/h at an altitude of 3500 m. Using dimensional analysis, estimate (*a*) its drag force and (*b*) the horsepower required to overcome air drag.

C5.1 Estimating pipe wall friction is one of the most common tasks in fluids engineering. For long circular, rough pipes in turbulent flow, wall shear τw is a function of density ρ, viscosity μ, average velocity V, pipe diameter d, and wall roughness height ε. Thus, functionally, we can write $\tau_w = \text{fcn}(\rho, \mu, V, d, \varepsilon)$. (a) Using dimensional analysis, rewrite this function in dimensionless form. (b) A certain pipe has $d = 5$ cm and $\varepsilon = 0.25$ mm. For flow of water at 20°C, measurements show the following values of wall shear stress:

Q (in gal/min)	~	1.5	3.0	6.0	9.0	12.0	14.0
τ_w (in Pa)	~	0.05	0.18	0.37	0.64	0.86	1.25

Plot this data in the dimensionless form suggested by your part (a) and suggest a curve-fit formula. Does your plot reveal the entire functional relation suggested in your part (a)?

P5.13 The speed of propagation C of a capillary wave in deep water is known to be a function only of density ρ, wavelength λ, and surface tension Υ. Find the proper functional relationship, completing it with a dimensionless constant. For a given density and wavelength, how does the propagation speed change if the surface tension is doubled?

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Practice Problem Solutions

P5.13 The speed of propagation C of a capillary (very small) wave in deep water is known to be a function only of density ρ, wavelength λ, and surface tension Y. Find the proper functional relationship, completing it with a dimensionless constant. For a given density and wavelength, how does the propagation speed change if the surface tension is doubled?

Solution: The "function" of ρ, λ, and Y must have velocity units. Thus

$$\{C\} = \{f(\rho, \lambda, Y)\}, \quad \text{or} \quad C = \text{const}\ \rho^a \lambda^b Y^c, \quad \text{or:} \quad \left\{\frac{L}{T}\right\} = \left\{\frac{M}{L^3}\right\}^a \{L\}^b \left\{\frac{M}{T^2}\right\}^c$$

$$\text{Solve for} \quad a = b = -1/2 \text{ and } c = +1/2, \quad \text{or:} \quad \mathbf{C = const}\ \sqrt{\frac{Y}{\rho\lambda}} \qquad \textit{Ans.}$$

Thus, for constant ρ and λ, if Y is doubled, C increases as $\sqrt{2}$, or **+41%**. *Ans.*

P5.14 Flow in a pipe is often measured with an orifice plate, as in Fig, P5.14. The volume flow Q is a function of the pressure drop Δp across the plate, the fluid density ρ, the pipe diameter D, and the orifice diameter d. Rewrite this functional relationship in dimensionless form.

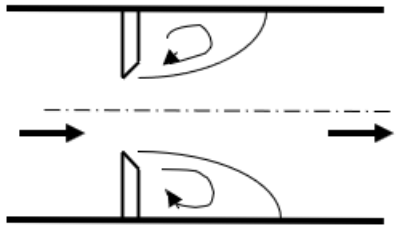


Fig. P5.14

Solution: Write out the dimensions of the variables:

Q	Δp	ρ	D	d
$\{L^3/T\}$	$\{M/LT^2\}$	$\{M/L^3\}$	$\{L\}$	$\{L\}$

There are five variables ($n = 5$) and three dimensions (MLT), so we expect 5-3 = 2 Pi groups. This can almost be done by inspection, although using (Δp, ρ,D) as repeating variables will also work fine. Only two, Q and Δp, contain time $\{T\}$, so we must divide $Q/\Delta p^{1/2}$, with dimensions $\{L^{7/2}/M^{1/2}\}$, to eliminate T. Then, to eliminate $\{M^{1/2}\}$, we multiply by $\rho^{1/2}$, giving $\rho Q/\Delta p^{1/2}$, with dimensions $\{L^2\}$. We finish by dividing by D^2. That is Π_1, and Π_2 is simply d/D. The final dimensionless function is

$$\frac{\rho Q}{D^2\sqrt{\Delta p}} = fcn\left(\frac{d}{D}\right) \qquad \textit{Ans.}$$

This is similar, but not identical, to the orifice plate function used in Chapter 6.

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Practice Problem Solutions

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V, mi/h:	20	40	60
Drag, lbf:	31	115	249

The same car travels in Colorado at 65 mi/h at an altitude of 3500 m. Using dimensional analysis, estimate (a) its drag force and (b) the horsepower required to overcome air drag.

Solution: For sea-level air in BG units, take $\rho \approx 0.00238$ slug/ft^3 and $\mu \approx 3.72E{-}7$ slug/ft·s. Convert the raw drag and velocity data into dimensionless form:

V (mi/hr):	20	40	60
CD = F/(ρV^{2}L^2):	0.237	0.220	0.211
ReL = ρVL/μ:	1.50E6	3.00E6	4.50E6

Drag coefficient plots versus Reynolds number in a very smooth fashion and is well fit (to ±1%) by the Power-law formula $\mathbf{C_D \approx 1.07 Re_L^{-0.106}}$.

(a) The new velocity is V = 65 mi/hr = 95.3 ft/s, and for air at 3500-m Standard Altitude (Table A-6) take $\rho = 0.001675$ slug/ft^3 and $\mu = 3.50E{-}7$ slug/ft·s. Then compute the new Reynolds number and use our Power-law above to estimate drag coefficient:

$$\mathbf{Re}_{Colorado} = \frac{\rho V L}{\mu} = \frac{(0.001675)(95.3)(8.0)}{3.50E{-}7} = 3.65E6, \quad \textit{hence}$$

$$C_D \approx \frac{1.07}{(3.65E6)^{0.106}} = 0.2157, \quad \therefore \ \mathbf{F} = 0.2157(0.001675)(95.3)^2(8.0)^2 = \mathbf{210\ lbf} \quad \textit{Ans.}\ (a)$$

(b) The horsepower required to overcome drag is

$$\text{Power} = FV = (210)(95.3) = 20030\ \text{ft·lbf/s} \div 550 = \mathbf{36.4\ hp} \quad \textit{Ans.}\ (b)$$

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Practice Problem Solutions

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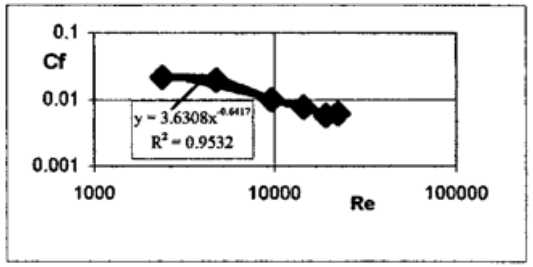
Solution: (a) There are 6 variables and 3 primary dimensions, therefore we expect 3 Pi groups. The traditional choices are:

$$\frac{\tau_w}{\rho V^2} = fcn\left(\frac{\rho V d}{\mu}, \frac{\varepsilon}{d}\right) \quad or: \quad \mathbf{C_f = fcn\left(Re, \frac{\varepsilon}{d}\right)} \quad Ans.\ (a)$$

(b) In nondimensionalizing and plotting the above data, we find that $\varepsilon/d = 0.25$ mm/50 mm = 0.005 *for all the data*. Therefore we only plot dimensionless shear versus Reynolds number, using $\rho = 998$ kg/m^3 and $\mu = 0.001$ kg/m·s for water. The results are tabulated as follows:

V, m/s	Re	Cf
0.0481972	2405	0.021567
0.0963944	4810	0.019411
0.1927888	9620	0.009975
0.2891832	14430	0.007668
0.3855776	19240	0.005796
0.4498406	22447	0.00619

When plotted on log-log paper, Cf versus Re makes a slightly curved line.
A reasonable power-law curve-fit is shown on the chart: $\mathbf{C_f \approx 3.63 Re^{-0.642}}$ with 95% correlation.
Ans. (b)
This curve is *only* for the narrow Reynolds number range 2000–22000 and a *single* ε/d.



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Thank You and Good Luck!



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